

Analysis Report

for

Online Retail Sales Performance

by:

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August 2026

GitHub Repository:
[ahmed-nassr/online-retail](https://github.com/ahmed-nassr/online-retail)

Dataset Link:
[Online Retail | Kaggle](https://www.kaggle.com/datasets/ahmed-nassr/online-retail)

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Executive Summary

This analysis examined the retailer's product sales between 1 December 2010 and 9 December 2011 to identify opportunities to increase gross product sales and reduce lost value through recorded cancellations. Non-product entries were excluded, such as discounts and bookkeeping adjustments. Customer-level results cover identified customers only because some transactions lack a CustomerID.

The following list is the total performance summary of the retailer from 1-12-2010 to 09-12-2011:

- Total Net Sales: £9,792,310
- Total Gross Sales: £10,271,035
- Total Cancelled Value: £478,724
- Total Number of Purchases: 19773
- Total Number of Cancellations: 3,422
- Average Order Value: £519.45
- Average Cancellation Value: £139.90
- Total Units Ordered (net): 5,304,966
- Total Units Ordered (gross): 5,576,996
- Total Units Cancelled: 272,030
- Cancellation Rate by Count: 17.31%
- Cancellation Rate by Value: 4.66%
- Cancellation Rate by Units: 4.88%
- Avg. Cancellation Value-to-Avg. Order Value: 26.93%
- Total Number of Identified Customers: 4,362
- Repeat-Purchase Rate for Identified-Customers: 65.27%
- Unique Products: 3,789

Sales performance was associated more strongly with units ordered than with product prices. Gross sales generally increased during the final months of 2011 and peaked near the end of the observed period, although December 2011 is incomplete and should not be compared directly with complete months. Purchase and cancellation counts generally moved in the same direction over time, while cancellation value was concentrated in several sharp episodes rather than consistently following sales activity.

Product sales were highly concentrated: the top 20% of products generated approximately 78.2% of gross product sales. **REGENCY CAKESTAND 3 TIER** led products by value, while **WHITE HANGING HEART T-LIGHT HOLDER** led by purchase frequency. **PAPER CRAFT, LITTLE BIRDIE** led by units because of a single exceptional order of approximately 80,000

units, demonstrating the need to distinguish products with broad demand from those whose performance is dominated by isolated wholesale transactions. Products observed across longer periods also accumulated greater sales activity.

Among identified customers, 65.27% placed at least two purchase orders and these repeat buyers generated 92.82% of identified-customer sales value. Customer value was also concentrated, with the top 20% of identified customers generating approximately 74.54%. These findings make repeat purchasing and high-value customer retention important areas for continued monitoring, although the results do not cover customers whose identifiers are missing.

Cancellation value was strongly influenced by cancelled unit volume and by a small number of extreme transactions: the two largest cancellation lines represented approximately half of total cancellation value. About 51.26% of purchased products and 35.39% of identified customers appeared in cancellation records. Because the dataset does not provide cancellation reasons or reliably connect every cancellation with its original purchase, the causes and full business meaning of these records cannot be determined from the available data alone.

Therefore, following the above conclusions, the following KPIs should be tracked to monitor the progress towards increasing and protecting net sales:

- Monthly Net Sales
- Monthly Gross Sales
- Monthly Purchase Count
- Monthly Total Gross Units Ordered
- Monthly Average Order Value
- Monthly Cancellation Count
- Monthly Cancellation Value
- Monthly Cancellation Units
- Monthly Cancellation Rate by Count
- Monthly Cancellation Rate by Value
- Monthly Cancellation Rate by Units
- Monthly Repeat-purchase Rate

Additional Recommendations:

- Separate broadly popular products from products whose performance is dominated by wholesalers.

- More investigation into the reasoning behind missing customers and how to reduce their occurrence is recommended for better results from the data.
- Develop retention plans for repeat-buyers.
- Monitor cancellations with extreme value separately from usual cancellations.

Objectives

The main objective of this analysis is to identify opportunities to increase and protect Net Product Sales by splitting the metric into two parts: Gross Sales, and Total Value Cancelled.

Methodology & Definitions

All money values are in sterling currency.

Cancellations were detected by having a C-prefixed invoice number and negative quantity.

Non-product entries such as accounting adjustments, discounts, and bookkeeping records were excluded.

The data has missing customers, so customer analysis contains only identified customers.

Metrics Definitions:

- Purchase/Order count: refers to the amount of total purchase activity.
- Cancellation: refers to the number of total cancellations.
- Gross Sales: Total value generated including orders that were later cancelled.
- Net Sales: Total value generated excluding orders that were later cancelled.
- Activity: generally, refers to value, purchases, and units ordered.
- Repeat-buyers: Customers that have bought at least 2 orders.

Overall Performance Snapshot

The following list is the total performance summary of the retailer from 1-12-2010 to 09-12-2011:

- Total Net Sales: 9,792,310
- Total Gross Sales: 10,271,035
- Total Cancelled Value: 478,724
- Total Number of Purchases: 19773
- Total Number of Cancellations: 3,422
- Average Order Value: 519.45
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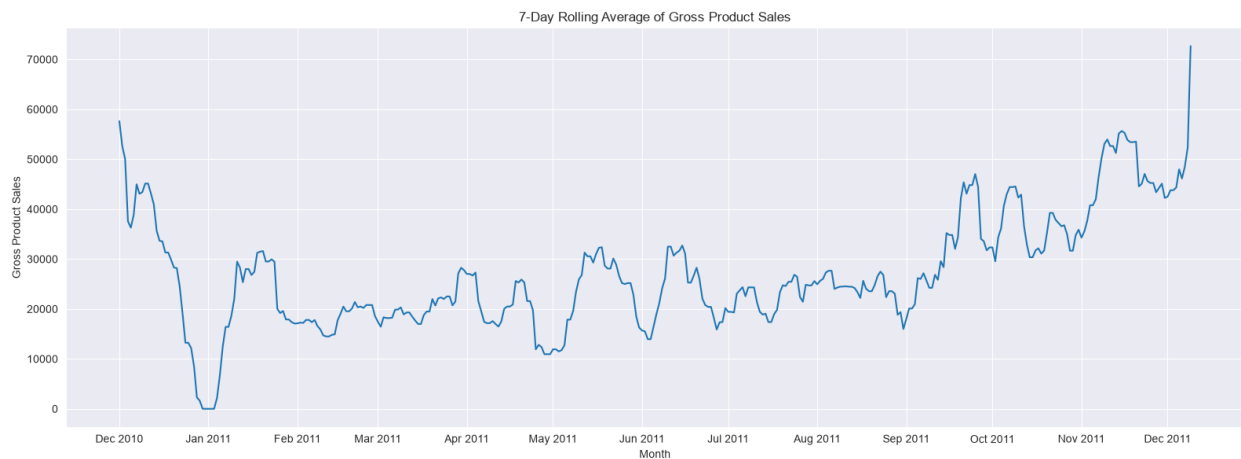
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General Sales Analysis

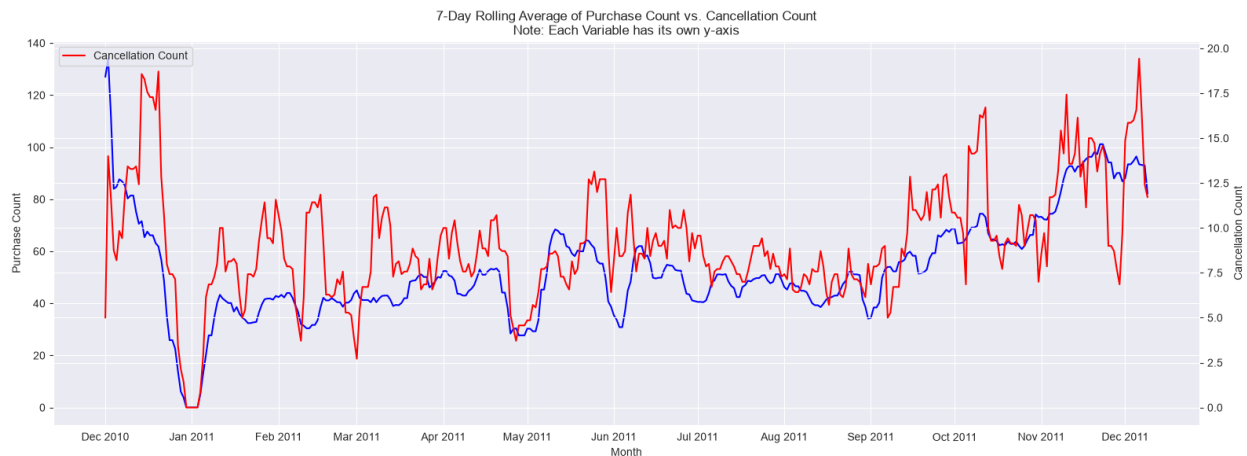
- The greatest driver of sales is number of units ordered.
- Product prices have very minimal effect over sales.
- 90.5% of orders are from the UK, the second most country with orders is Germany which has only 2.2% of orders.
- High-value-generating orders tend to have higher total product units ordered.

Time trends

7-Day Rolling Average of Gross Product Sales:



7-Day Rolling Average of Purchase Count vs. Cancellation Count (y-axis is NOT matched):



- Data spans across from 01-12-2010 to 09-12-2011 (DD-MM-YYYY format).
- Days with more units bought tend to have higher value per order.

- There was a significant drop in gross sales at the end of the year 2010, reaching zero sales, purchases, and cancellations. At first, it looks like it was caused mainly by a significant lack of purchase count, but zero purchase count is unlikely to have been caused by customer behavior and is more likely to be things like maintenance or system errors. Though, without stakeholders' guidance or access to system records, it's not possible to know the cause for sure.
- Gross sales went up in the holiday season, peaking at the start of December 2011.
- Even though December 2011 only spans over 8 days compared to 20 days in December 2010, Sales in December 2011 are not that much less than in December 2010, which carries the potential meaning of the company's improvement after 1 year.

Product Analysis

Histogram of Purchase Frequency Across Products:



- The pricing of a product does not necessarily affect product activity.
- More expensive products tend to be associated with less units ordered.
- The average product lifespan is 276 days. Over 50% of products have over 350 days of lifespans.
- Products with longer lifespans generally drive more activity.
- The top 20% of products represent ~78.2% of total value, and the rest of the 80% represent 21.8% of total value. This shows that values of products follow the Pareto principle.

Product Rankings

Top 10 Products by Value:

1. REGENCY CAKESTAND 3 TIER
2. PAPER CRAFT , LITTLE BIRDIE (only 1 purchase with 80k units)
3. WHITE HANGING HEART T-LIGHT HOLDER
4. PARTY BUNTING
5. JUMBO BAG RED RETROSPOT
6. MEDIUM CERAMIC TOP STORAGE JAR
7. RABBIT NIGHT LIGHT
8. PAPER CHAIN KIT 50'S CHRISTMAS
9. ASSORTED COLOUR BIRD ORNAMENT

10. CHILLI LIGHTS

Top 10 Products by Purchase Count:

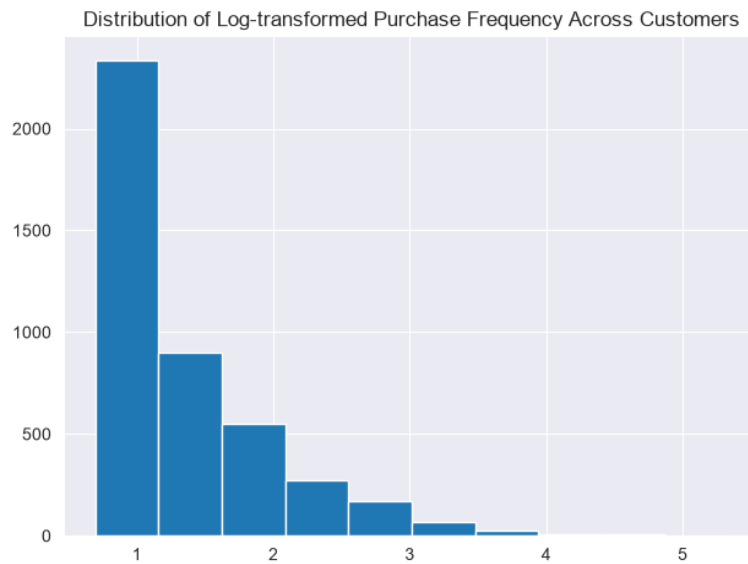
1. WHITE HANGING HEART T-LIGHT HOLDER
2. JUMBO BAG RED RETROSPOT
3. REGENCY CAKESTAND 3 TIER
4. PARTY BUNTING
5. LUNCH BAG RED RETROSPOT
6. ASSORTED COLOUR BIRD ORNAMENT
7. POPCORN HOLDER
8. SET OF 3 CAKE TINS PANTRY DESIGN
9. PACK OF 72 RETROSPOT CAKE CASES
10. LUNCH BAG SUKI DESIGN

Top 10 Products by Units:

1. PAPER CRAFT , LITTLE BIRDIE (only 1 purchase with 80k units)
2. MEDIUM CERAMIC TOP STORAGE JAR
3. POPCORN HOLDER
4. WORLD WAR 2 GLIDERS ASSTD DESIGNS
5. JUMBO BAG RED RETROSPOT
6. WHITE HANGING HEART T-LIGHT HOLDER
7. ASSORTED COLOUR BIRD ORNAMENT
8. PACK OF 72 RETROSPOT CAKE CASES
9. RABBIT NIGHT LIGHT
10. MINI PAINT SET VINTAGE

Identified-Customer Analysis

Histogram of Log-transformed Purchase Frequency Across Customers:



- The distribution of log-transformed purchase frequency is right-skewed, meaning that the original distribution is even more right-skewed. This implies that most customers order at low frequency.
- The repeat-purchase rate is: 65.27%. This indicates outstanding customer loyalty.
- Despite representing 65.27% of customers, repeat-buyers represent 92.82% of total customer value.
- The average customer lifespan is 130 days. Although most customers have been active recently, their observed lifespans remain relatively short because most customers made their first purchase on a recent date.
- Customers with high lifespans tend to drive more activity.
- The top 20% of identified customers represent ~74.54% of total value, and the rest of the 80% represent 25.46% of total value. This shows that customers follow the Pareto principle.

Cancellation Analysis

- Cancelled value is mainly driven by cancelled units.
- The top 2 cancellation transactions (rows in cancellations) represent about half of total cancellation value.
- Average cancellation value is ~73% less than average order value, which means that cancellations mostly happen with low-value orders.
- While cancellation count generally follows purchase count, most cancellation value is concentrated in few sharp episodes, mainly in January, April, October, and the final recorded days in December 2011.
- Only 51.26% of unique products are present in cancellations.
- Only 35.39% of unique customers contribute to cancellations.
- Cancellations are not mainly driven by certain products or customers.
- Cancellations are not mainly associated with high-value or low-value products or customers.

Conclusion & Recommendations

Net sales are broken down into two parts: gross sales, and cancellation value. Additionally, the greatest driver of both gross sales and cancellation value is units ordered/cancelled. The highest performing category of customers is the repeat-buyers, contributing to 92.82% of gross sales generated. The products and customers with higher lifespans tend to drive higher overall activity.

Therefore, following the above conclusions, the following KPIs should be tracked to monitor the progress towards increasing and protecting net sales:

KPIs

- Monthly Net Sales
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Additional Recommendations

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- Develop retention plans for repeat-buyers.
- Monitor cancellations with extreme value separately from usual cancellations.